Unique prime input partition simultaneously factors stable-matching execution and lattice structures

Yoshiteru Ishida

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Editorial note. One principal unique-finest-partition/simultaneous-product object, with both original Theorem6 and Theorem7 statements. Complete original mathematical document85–1956, all source-local proofs, figures and embedded numbered bibliography retained. Original proofs are paragraph-form Pandoc prose; no proof environment or mathematical argument is invented.

Editorial note. Original PROVED/EXHAUSTIVE/TARGETED annotations, independent implementation statements and census values are author statements retained as context, not agent certifications or new counts. Ancillary scripts/README are preserved verbatim as source assets without execution, verification or relicensing.

Editorial note. Original1795–1801 Generative AI disclosure names ChatGPT/Gemini/Claude for organizing claims, language editing and proposing/running verification checks; author states every mathematical statement/proof/computation/citation was reviewed and takes responsibility. Literal disclosure retained without selected-core discovery inference, AI-free labeling or AI-dependency investigation.

Editorial note. Authors retain responsibility for mathematics. No proof repair, new assumptions, independent proof review, counterexample search or certification.

Editorial note. Actual r001 runaway hypertarget caused by assembly boundary cutting original lines250–252. Move the context/proof boundary one line earlier, retaining the complete original command and every body line exactly once. No original mathematical or prose text changes.

Allowed standard prerequisites (editorial).

- Stable marriage with strict complete preferences, reachable Gale-Shapley events, antimatroid feasible prefixes and stable-matching lattice order. Native finite balanced instance, strict complete preference lists and input-independent block definitions retained. Author distinguishes finest input blocks from mere decompositions and stable-set decompositions; original prior-attribution discussion retained.. Complete original laminarity/closed-cell and simultaneous-product arguments included without reconstructing cited stable-matching foundations or replaying ancillary census checks.

Abstract

Every balanced instance of the stable marriage problem with strict complete preferences has a unique finest partition into prime blocks, and that single partition simultaneously factors three different

structures: the reachable execution digraph as a Cartesian product, the proposal-prefix antimatroid as a direct sum, and the stable-matching lattice as a direct product. The converse fails, and fails at every size from two on: two explicit families share the identical Boolean-cube execution while one is maximally decomposable with a single stable matching and the other is prime with n. Uniqueness yields an exact census, a recursion counting the prime instances at every size, under which exactly 88,478,208 of the 110,075,314,176 profiles with four agents on each side are decomposable and the decomposable fraction is asymptotically $n! / n^{2n}$. The blocks are characterised as the square components of the mutual-rank filtration, so the partition is computable in polynomial time and the factorisation is a tool rather than only a fact.

1. Introduction

Deferred acceptance is normally analysed through its terminal stable matching. This paper analyses the instance instead, and asks the question one asks of an integer: can it be broken into smaller pieces, is the breaking unique, and what is preserved by it.

The answer is that it can, that it is, and that a surprising amount is preserved. An instance decomposes when its agents split into paired blocks of proposers and receivers, each block mutually top-ranked within itself. Such a partition always has a unique finest refinement (Theorem 6), and the blocks of that refinement are the prime instances of the title. What makes the decomposition worth having is not the uniqueness alone but what one partition does at once (Theorem 7): the execution digraph becomes a Cartesian product over the blocks, the antimatroid of feasible proposal prefixes becomes a direct sum, and the lattice of stable matchings becomes a direct product. Three structures that are otherwise studied separately are factored by the same object.

That makes the theorem a reduction tool. Any quantity that composes over a product - the state count, the proposal total, the number of stable matchings, the commutation index - is determined by its values on prime blocks, and Corollary 7.2 sorts those quantities into the multiplicative ones and the additive ones. To compute such a quantity it suffices to find the finest partition, which Theorem 9.4 reduces to reading off the square components of a filtration of bipartite graphs, and this can be done in polynomial time.

The limits of the tool are as sharp as the tool. Behavioural independence - the execution splitting as a direct sum - does not imply input independence (Theorem 8), and the failure is not exceptional: Proposition 12.3 exhibits two families, one at every size, that share the identical Boolean-cube execution while sitting at opposite ends of both of the things the execution cannot see. Section 5 makes that precise, and Proposition 9.10 shows that the behavioural decomposition, unlike the input one, carries no nesting at all: its factorisations form an interval in the partition lattice and their number is a Bell number depending on nothing but how many prime factors there are.

Finally, uniqueness is what makes counting possible. Theorem 9 turns the finest partition into a recursion for the number of prime instances at every size, with exact values at small sizes and an asymptotic decomposable fraction of $n! / n^{2n}$. The small values are given in full so that a reader can check an independent implementation against them; that is the part of a structure theory that makes it usable by other people.

Related work. Deferred acceptance and the lattice of stable matchings are classical [1,2,3]. The structure of what an algorithm is able to do next is the greedoid tradition of Korte, Lovasz and Schrader [4]; antimatroids induced by matchings are studied by Kawase and Yamaguchi [5], and

which preference profiles determine the proposals the Gale-Shapley algorithm makes is the question of Sukegawa and Yamamoto [6]. A uniqueness statement of the kind proved here was first isolated for lattices by Dilworth [7] and for graphs under the Cartesian product by Sabidussi [8], with later extensions to graphs with loops [9]; convex geometries, the duals of antimatroids, are due to Edelman and Jamison [10]. That independent proposals commute is the phenomenon trace theory [11] and Keller's parallel program schemata [12] axiomatise, and Section 4 says exactly how far commutation alone goes. Bounds on the number of stable matchings are due to Karlin, Oveis Gharan and Weber [13] from above, to Benjamin, Converse and Krieger [14] from below, and at small sizes to Thurber [15]. Order-independence of the kind Section 2 proves is the defining property of abelian networks [16] and was established for threshold dynamics by Goles and Olivos [17]. Closest to the subject of this paper, the decomposability of a single instance was studied by Veklerov [18], whose first criterion is the static condition used here, and the Latin instances - those in which the two ranks of a pair sum to n + 1 - go back to Knuth [19] and are the latin marriages of [14]. The paper continues the author's [20], which gives the operator formulation of the run used in Section 2, and [21], whose regular anti-phase templates supply the two families of Section 5.

Relation to the author's monograph. Some of the material below was presented without proof, through worked examples rather than statements, in the author's monograph [22]. Its Chapter 3 classifies the size-three instances by diagrams and introduces the decomposition of an instance informally, Chapter 4 studies the Latin instances of [19,14], Chapter 5 introduces a cyclic family with two stable matchings, and Chapter 7 states the block condition used here in the form of [18]; the size-three classification of Chapter 3 first appeared in [23]. The monograph proves none of it. Its Chapter 1 names the gap directly - "for the classification and efficient solving of the stable marriage problem, characterization of the conditions for decomposing instances of SMP to non-decomposable ones is critical. Although several versions of decompositions of SMP are possible, their characterization has not yet been completed" - and its Section 3.5 adds that the decomposition of an instance need not be unique. This paper closes that gap for one precise sense of the word: a finest partition into input-independent blocks always exists and is unique (Theorem 6), the blocks are the square components of a filtration and so are computable (Theorem 9.4), they form a partition exactly when the maximal ones cover (Theorem 9.6), and the instances admitting none can be counted at every size (Theorem 9). Apart from the sentence quoted above, no text, figure, table or numerical value is taken from the monograph; the figure below is generated from the definitions by the accompanying code.

Recent work in other directions. Deferred acceptance continues to be studied from several directions that this paper does not touch: agents who misreport or mistake their own preferences [24], matroid and partial-order constraints on what counts as a matching [25], preferences that change after the fact [26,27], distributed and truthful implementations [28], the running time of the algorithm itself [29], lattice-linear predicate methods for constrained variants with ties [30], and local search when ties are broken [31]. Small instances have also been catalogued before, as integer sequences [32] and in the OEIS [33]. The present paper is orthogonal to all of these: it is about the structure of a single instance and what that structure does to the algorithm run on it, and it stays inside strict complete preferences throughout. Two further preprints of the author study the same instance space from other angles [34,35].

How the statements are labelled. Every numbered statement carries a tag saying how it is established, because this paper mixes proofs with complete computations and the difference matters. PROVED means a proof is given here or cited. EXHAUSTIVE means the claim was checked on every instance

of the size named; at size three that is all 46,656 profiles with strict complete preferences on both sides. EXACT means a finite candidate set was fixed in advance by a proved statement and every candidate in it was then settled, which is how sizes four to six are reached where enumerating instances is out of the question. TARGETED means a particular instance was constructed or searched for and is supplied as a certificate. A tag never promotes a computation to a proof: where a statement holds at every size the tag says PROVED, and where it was verified only at certain sizes the tag names them.

Organisation. Section 2 fixes the execution model and proves the two facts the rest depends on: a reachable state is its pointer vector, and the reachable set is scheduler-independent. Section 3 defines the blocks, proves that a finest partition into them exists and is unique, and characterises them as the square components of a filtration. Section 4 is the products theorem and the composition laws. Section 5 shows that the converse fails and how far. Section 6 is the census, its asymptotics and the recursion behind it. Section 7 lists five open problems. Two companion papers are referred to: one on the Latin-rank-balance setting, and a computational classification of the execution structure at small sizes, in preparation, which is where the atlas of size-three executions and the enumerable quotient at sizes five and six are developed.

2. Preliminaries: the reachable event system

Model and reachable event system

Let P = {p_0, …, p_(n-1)} be proposers and R = {r_0, …, r_(n-1)} receivers. Unless stated otherwise every agent has a strict complete order over the opposite side; such an instance is balanced and complete. A state is s = (h, q), where h(r) is the proposer currently held by r or the empty symbol, and q(p) is the number of proposals p has already made. The canonical initial state has h empty and q = 0. A proposer is active at s when it is unheld and still has an unproposed receiver. The local update U_p advances q(p) by one, sends the proposal to the next receiver on p's list, and lets that receiver retain the better of its current holder and the new proposer. A scheduler repeatedly picks an active proposer.

The event e = (p, i) denotes p making its (i+1)-st proposal. For a pointer vector q the corresponding event set is F(q) = {(p,i) : 0 <= i < q(p)}. The rooted reachable digraph G(I) contains every state reachable from the initial state under every legal serial schedule; edges point away from the root and are coloured by proposer. We also use the uncoloured digraph and the underlying undirected graph.

A family of subsets of a finite event set is an antimatroid when it contains the empty set, is accessible, and is closed under union. A poset antimatroid is the family of down-sets of a partial order; for the finite families here this is equivalent to being closed under intersection as well. Intersection closure was tested separately from the antimatroid axioms.

Two notions of independence

Input-independent partition (static). Paired blocks P = P_1 + … + P_s and R = R_1 + … + R_s with |P_i| = |R_i| > 0, such that every proposer in P_i ranks all of R_i above every receiver outside R_i, and every receiver in R_i ranks all of P_i above every proposer outside P_i. Equivalently, for each i the top |R_i| receivers of each p in P_i are exactly R_i, and symmetrically on the other side. An instance is input-prime when no partition with s >= 2 exists. Candidate blocks were generated

from mutual top-k coalitions and assembled by exact cover, because a mutual top coalition need not have an independent complement.

Behavioural decomposition (dynamic). Let E = E_1 + … + E_t partition the proposal events that actually occur, and set F_i = {X intersect E_i : X in F}. The event system is behaviourally decomposable when F is the set of all disjoint unions X_1 + … + X_t with X_i in F_i. This preserves event identities and proposal chains; it is strictly stronger than factoring the underlying uncoloured undirected graph, and it is distinct from input independence. Operationally, two proposers of one behavioural block are joined when they propose to a common receiver during the run. One direction is immediate (proposers that never share a receiver address disjoint receiver sets and run independently), while the converse, that the connected components of that graph are exactly the behavioural blocks, is verified rather than proved; Section 4 states what was checked. The graph is well defined because the event set is schedule-independent, and the statements below rest on it only through the direction that is proved, or at sizes where the two tests were both computed and agree.

A third notion, and where the static one comes from. A third sense of the word is in the literature. Veklerov [18] calls an instance decomposable when every stable matching pairs a fixed set of proposers with a fixed set of receivers and with no one else, which is a condition on the stable set rather than on the preferences or on the run. Input independence implies it, by Theorem 7, but behavioural decomposition does not. Take n = 2 with p_1 preferring r_1 and p_2 preferring r_2, while r_1 prefers p_2 and r_2 prefers p_1. Each proposer makes one proposal and is accepted, so the event system splits into two single-event factors; but both perfect matchings are stable and they disagree on both proposers, so no proper set of proposers is paired with a fixed set of receivers. The third sense is implied by the static notion and incomparable with the dynamic one. He gives three sufficient criteria and one necessary one, and observes that a decomposition in his sense need not be unique: an instance with a single stable matching has every splitting of that matching as a decomposition. His first criterion is the static condition of this paper, stated in the same words - a set of proposers and an equinumerous set of receivers, each of which ranks the other set above every outsider - and his second is a one-sided form of it; [22, Chapter 7] restates both through the top-k operator and attributes them to [18]. Neither is developed into a partition in either place. The third sense does have a finest partition, and a unique one: a partition is valid for it exactly when no edge of the union of all stable matchings crosses it, so the valid partitions are the unions of connected components of that union; each component is balanced, because any stable matching restricts to a perfect matching on it, and the component partition is therefore the finest. Both partitions are functions of the input, and both can be computed without enumerating anything exponential, since the pairs that lie in some stable matching are found by the rotation machinery of [2], which goes back to Irving and Leather [36]. The difference is what the condition is about. Veklerov's is a condition on the stable set, and it factors the stable set, which is what it is about. The partition below is of the other kind: the blocks are laminar, so a finest partition into them exists, is unique (Theorem 6) and is determined by the ranked input alone; they are the square components of a filtration (Theorem 9.4), so the partition is computed from the preferences without the stable set; they form a partition exactly when the maximal ones cover (Theorem 9.6); and the instances admitting no such partition can be counted at every size (Theorem 9). Theorem 8 separates the first two notions; the relation of the third to the run is not pursued here.

Two earlier papers of the author's belong to this lineage, and naming how they are weaker is the clearest way to say what is new. [37] shows that when all proposers share one preference order the

instance reduces to n mutual-first pairs removed one at a time, so that its stable set is the direct product of the stable sets of the pieces. That is the singleton case of Theorem 7 for the stable set alone, and the reduction is sequential: a pair is mutually first only after the earlier pairs have been taken out, so it does not exhibit a partition of the original input. The aligned family A_n of Section 5 is not that situation and is worth contrasting with it: there the proposers' first choices are distinct and already mutual, so the singleton partition is input-independent from the start, whereas under a shared order the first choices coincide and a pair becomes mutually first only once the earlier pairs are gone. The same paper calls two proposers k-indistinguishable when their first k choices agree, observes that the classes refine as k grows, and asks how deep into the lists one must look; the mutual-rank filtration of Section 3 is the two-sided form of that question, and it is mutuality rather than depth that makes the blocks laminar and the finest partition unique. [38] takes the decomposition from the blocking diagram instead, calls a top connected component indecomposable, and observes that Veklerov's condition is its proposer-oriented case. Both are one-sided, and both read the decomposition off the stable set or the run rather than off the ranked input.

The operator formulation of the run - a local update for each active proposer, a scheduler choosing among them, and the progress invariants that give termination, stability, proposer optimality and schedule independence - is not reproved here. It is the subject of a companion preprint and is cited as such. What is new below is the structure of the FAMILY of reachable states, which that preprint does not study.

Lemma 1 (states are determined by proposal counts) [PROVED]

At every reachable state the proposal-count vector q determines the held-proposer vector h.

Proof. For each proposer, q(p) fixes the initial segment of receivers already contacted, so the multiset of past proposals is reconstructed from q alone. Each receiver retains its most preferred proposer among those it has received, so applying each receiver's order to that reconstructed multiset yields h. No assumption of completeness or balance is used. QED

The reachable graph can therefore be stored without a matching coordinate, and q-injectivity holds: distinct reachable states have distinct pointer vectors.

Theorem 2 (local commutativity) [PROVED]

If two distinct proposers are active at a reachable state, executing their next proposals in either order gives the same state after both.

Proof. Each pointer advances once under both orders. If the two proposals target different receivers, the two holder updates are disjoint. If they target the same receiver, the final holder is that receiver's maximum among its previous holder and the two new proposers, which does not depend on arrival order; the same agents are rejected either way. QED

Lemma 3 (persistent enablement) [PROVED]

If an event is enabled at s and a different proposer acts, the event remains enabled until its own proposer executes it.

Proof. An active proposer is unheld and has a next receiver on its list. No other proposer's action can make it held, and no other proposer consumes its next proposal. QED

Theorem 4 (antimatroid and exchange) [PROVED]

The family F(I) of feasible proposal-prefix event sets is an antimatroid, and any two complete executions with the same event set are connected by a sequence of adjacent exchanges of events belonging to different proposers.

Proof. The empty set is feasible. Accessibility follows by deleting the last event of any non-empty feasible prefix. For union closure, take realisations of two feasible sets, start from the first, and append the events present only in the second in their proposer-chain order, skipping events already present; by Lemma 3 each required next event is still enabled when reached. The same exchange argument moves a chosen enabled event leftward through a complete execution, and repetition connects any two complete executions with the same event set. QED

Corollary 4.1. The feasible-set lattice is join-distributive. Its join is set union, equivalently coordinatewise maximum of pointer vectors. The meet need not be set intersection.

Corollary 4.2. F(I) is a poset antimatroid, and the lattice is the distributive lattice of down-sets, exactly when F(I) is also intersection-closed.

Intersection closure is common but not universal. EXHAUSTIVE at n = 3: 41,472 of the 46,656 profiles give poset antimatroids and 5,184 do not. Of the 20,736 behaviourally prime profiles, 15,552 are poset and 5,184 are not; so behavioural primeness does not imply non-poset structure, and every non-poset profile is behaviourally prime.

Caution on trace theory. The exchange statement of Theorem 4 resembles Mazurkiewicz trace equivalence, but the independence relation here is state-dependent: two events are independent at some states and not at others, because enablement depends on earlier rejections. The system is therefore not a trace monoid over a fixed independence alphabet; it is a concurrent transition system with the strong-confluence property, and the attribution in the discussion is to that weaker frame [11,12]. Local commutativity alone is not sufficient for global confluence; the argument above uses persistent enablement as well.

Theorem 5 (Boolean-cube characterisation) [PROVED]

For a balanced instance with strict complete preferences, G(I) is the n-dimensional Boolean cube if and only if the proposers' first choices are pairwise distinct.

Proof. If the first choices are distinct no proposal is ever rejected, so each proposer has made either no proposal or exactly one, every subset of first proposals is feasible, and the reachable graph is the cube. Conversely, a first-choice collision in a complete balanced market produces a rejected proposer that still has a proposal available, hence a proposal chain longer than one coordinate. QED

Corollary 5.1. EXHAUSTIVE at n = 3: 10,368 of 46,656 profiles are Boolean cubes. The count is also analytic: 48 of the 216 proposer halves have distinct first choices, and the receiver half is unconstrained, giving 48 x 216.

The converse fails outside the hypotheses. With incomplete lists the cube can occur with a collision: two proposers whose lists are the single receiver r_0, with r_0 ranking them in either order, give the Boolean cube B_2 although both first choices coincide. The unbalanced instance with two proposers and one receiver behaves the same way. Completeness and balance are therefore essential, not cosmetic. [PROVED, with explicit certificates]

3. Blocks, and the unique finest partition

Theorem 6 (uniqueness of the finest input partition) [PROVED]

Every finite balanced instance with strict complete preferences has a unique finest input-independent partition.

Proof. Let (A,B) and (C,D) be blocks from two input-independent partitions. Suppose A and C overlap and let p lie in the overlap. If neither B nor D contains the other, pick b in B minus D and d in D minus B. Membership of p in A forces b above d, and membership of p in C forces d above b, contradicting strictness. Hence B and D are nested, and the receiver-side argument nests A and C the same way. If A and C are disjoint then so are B and D. The paired blocks of any two partitions are therefore laminar, so the inclusion-minimal blocks of their union form a common refinement that is again input-independent. Iterating over the finite set of partitions gives a unique finest one. QED

A useful consequence used repeatedly below: the finest partition of an instance is the union of the finest partitions of its blocks, and a block's restriction is input-prime exactly when that block cannot be refined. The uniqueness is worth stating explicitly because the literature records the decomposition of an instance as ambiguous. Veklerov [18] exhibits an instance whose single stable matching admits decompositions into two, three and four parts at once, and [22, Section 3.5] repeats that the decomposition of an instance need not be unique. That ambiguity is about decompositions in general and not about finest ones, and taking the finest removes it in either sense of the word, Veklerov's finest being the partition into connected components of the union of the stable matchings. What Theorem 6 adds is that the finest partition into input-independent blocks exists, is unique, and is a function of the ranked input, so it is in hand before any stable matching is computed.

Uniqueness on its own does not say how to find the partition, or how to tell whether a given instance has one. Both follow from a graph description of the blocks.

The structure of input-prime instances

The blocks have a graph description that makes primality easy to see and easy to test.

The mutual-rank filtration. For 1 <= k <= n let M_k be the bipartite graph on P together with R that joins p to r when r is among the top k of p and p is among the top k of r. These nest,

M_1 contained in M_2 contained in … contained in M_n = K_(n,n),

and every vertex of M_k has degree at most k. M_1 is the graph of mutual-first pairs, a partial matching; the filtration ends at the complete bipartite graph. Call this the mutual-rank filtration of I. Its one-sided ancestor is the k-indistinguishability of [37], which partitions the proposers alone by agreement of their first k choices; the classes there also refine as k grows, but they pair with no receivers, and it is the mutual edge that turns a nested family of equivalence relations into blocks that are laminar on both sides.

Theorem 9.4 (blocks are the square components of the filtration). [PROVED; EXHAUSTIVE at n = 3, verified at n = 4] A pair (A, B) with |A| = |B| = k is an input-independent block if and only if the union of A and B is a connected component of M_k isomorphic to K_(k,k), equivalently a component of M_k in which every vertex has the maximum degree k.

Proof. Let (A, B) be a block. Each p in A has top k equal to B, and each r in B has top k equal

to A, so every pair in A times B is an edge of M_k and the induced subgraph contains K_(k,k). Since p has only k receivers of rank at most k and all lie in B, p has no M_k-edge outside B, and symmetrically for receivers; so A together with B is a full component and the induced graph is exactly K_(k,k). Conversely, if a component is a copy of K_(k,k) on A and B with |A| = |B| = k, then each p in A has k mutual edges, all into B, and p has at most k receivers of rank at most k, so its top k is exactly B; symmetrically on the other side. QED

Proposition 9.5 (the blocks form a forest). [PROVED] Any two blocks are nested or disjoint on each side, so the blocks of I, ordered by inclusion and rooted at the trivial block, form a forest.

Proof. This is the laminarity argument of Theorem 6, which never used the assumption that the blocks belong to partitions: if p lies in A intersect C with top |B| equal to B and top |D| equal to D, then an element of B outside D and an element of D outside B would have to be ranked above one another both ways. QED

Theorem 9.6 (primality is a covering failure). [PROVED; EXHAUSTIVE at n = 3] I is input-decomposable if and only if the inclusion-maximal proper blocks cover P, equivalently cover R. Having a proper block is not sufficient.

Proof. The maximal proper blocks are pairwise disjoint by Proposition 9.5, so if they cover they form a partition, and each is input-independent by definition. Conversely any partition into proper blocks consists of blocks contained in maximal ones, which therefore cover. QED

The last sentence of the theorem is the point, and it is what makes primality a subtle property rather than an obvious one. EXHAUSTIVE at n = 3: 36,360 of the 46,656 instances, a little over seven in every nine, possess a proper block and are nevertheless input-prime; the same holds for 14,914 of 20,000 random size-four instances. The smallest witness has n = 2 and one mutual-first pair: the pair is a block, the remaining pair is not, and nothing covers, so a two-agent instance can carry a block and stay prime. An input-prime instance is therefore not one without local structure; it is one whose local structure fails to tile.

Proposition 9.7 (the exact-cover polynomial, and why it factorises). [PROVED; EXHAUSTIVE at n = 3; TARGETED at n = 4 to 6] Give every agent an arc to each of its top k choices on the other side, and call a set of k proposers and k receivers a closed k-cell when every such arc from a member of the set stays inside it. Then:

(a) the closed k-cells are exactly the blocks of Theorem 9.4;

(b) writing p(U) for the least proposer of a balanced pair (U, V), the recursion H(empty, empty) = 1 and H(U, V) = the sum, over closed cells (A, B) with p(U) in A inside U and B inside V, of z times H(U less A, V less B) has the property that the coefficient of z^r in H(P, R) is the number of exact covers of the agents by r blocks; so I is input-prime exactly when H(P, R) = z;

(c) H factorises over the block forest. If the maximal proper blocks of a block b partition b, then H_b = z + the product of H_c over those blocks; otherwise H_b = z.

Proof. (a) If (A, B) is a block of size k then every member of A ranks exactly B in its first k places and every member of B ranks exactly A, so no arc leaves the set. Conversely, closure of a balanced set of 2k agents forces the k arcs of each member to exhaust the opposite side of the set, which is the block condition. Closure therefore delivers the mutual condition of Theorem 9.4 by itself, which is why the directed form needs no reciprocity hypothesis and is the shorter of the two definitions. (b)

Distinguishing the least proposer counts each cover once, because exactly one cell of a cover contains it, and deleting that cell leaves an exact cover of the balanced remainder; the converse adjoins one eligible cell to such a cover. (c) Blocks are laminar by Proposition 9.5, so a block properly inside b lies inside one of the maximal proper blocks of b, and those are pairwise disjoint. If they fail to cover b, some agent of b lies in no proper block, so no cover by proper cells exists and H_b = z, which is Theorem 9.6. If they do cover, every cover other than b itself restricts to an exact cover of each of them, and any combination of such covers recombines into one of b, which is the product. QED

Three things follow, and they are the reason for stating the polynomial rather than only checking it. First, part (c) turns the recursion of part (b) from a search over cells into a walk over the block forest, linear in the size of that forest. Second, the degree of H is the number of blocks in the finest partition and the leading coefficient is 1; that coefficient is Theorem 6 read off the polynomial, since a second finest factorisation would contribute a second monomial of top degree. Third, H evaluated at 1, less 1, counts the non-trivial input-independent partitions. One object therefore carries Theorem 6, Theorem 9.4 and Theorem 9.6 at once, and this is the Hall-shaped recursive form that Proposition 21 was reaching for: Hall's own induction separates a tight set from the residual problem, and here the distinguished proposer separates one closed cell from a balanced remainder.

Table 17. The size-three census by exact-cover polynomial. [EXHAUSTIVE] Every one of the 46,656 profiles, classified by its polynomial and by the two partitions the polynomial reconciles: the coarsest non-trivial one, given by the maximal proper blocks, and the finest one.

exact-cover polynomial	maximal proper blocks	finest partition	profiles
z	none that cover	prime	45,768
$z + z^2$	1+2	1+2	504
$z + z^3$	1+1+1	1+1+1	312
$z + z^2 + z^3$	1+2	1+1+1	72

The last row is the one to read. In 72 of the 888 decomposable size-three instances the coarsest non-trivial partition and the finest one differ: the two-agent block of the coarsening is itself decomposable into singletons. The polynomial sees that and the shape of the maximal blocks alone does not, which is worth keeping straight, because the shape counts quoted elsewhere in this manuscript, 504 and 384, count finest partitions; counted by maximal blocks the same profiles split 576 and 312. A second 72 sits nearby, the profiles with a non-trivial input automorphism, and the two sets are disjoint; the coincidence of size is only that, since an automorphism of order three cannot fix a partition of shape 1+2.

Verification. Over all 46,656 size-three profiles, and over 5,500 random and 9,500 planted decomposable profiles at n = 4, 5 and 6, six things were checked and none failed: the closed cells agreed with the blocks of Theorem 9.4; the coefficients of H agreed with a brute-force enumeration of every exact cover; the criterion H = z agreed with Theorem 9.6; the forest factorisation of part (c) agreed with the recursion of part (b), coefficient by coefficient; the leading coefficient was 1; and the degree was the number of blocks in the finest partition. The coefficients are not all 0 or 1, which is the sign that the object is doing work: a size-four instance with finest partition 1+1+1+1 and two disjoint two-agent blocks has polynomial $z + z^2 + 2z^3 + z^4$, the 2 counting the two ways to split one of the pairs and not the other.

4. One partition, three products

This is the section the paper exists for. The partition of Section 3 was defined by a condition on the ranked input alone, and there is no reason from the definition to expect it to say anything about the run or about the stable set. It says everything about both.

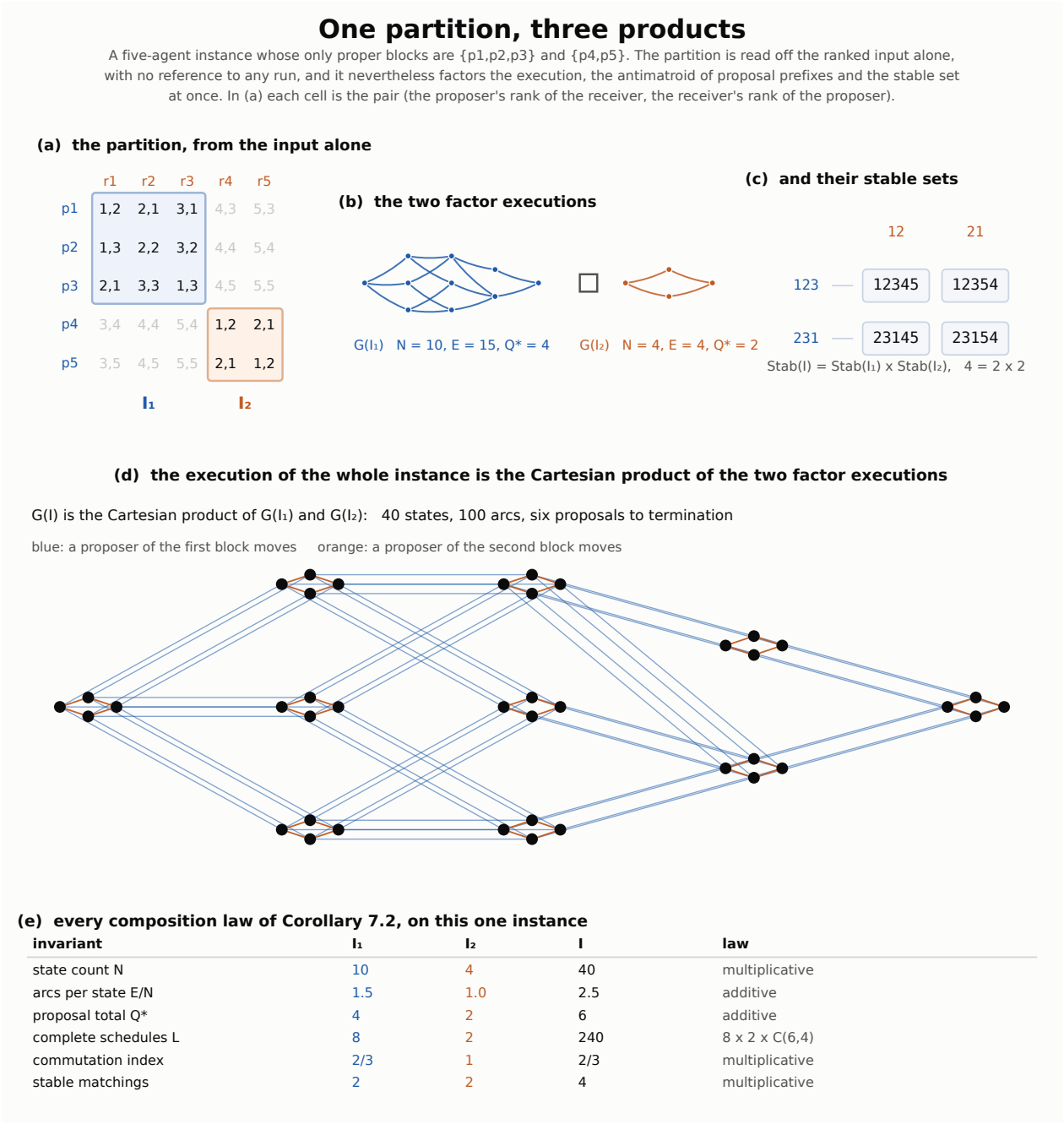
Theorem 7 (products) [PROVED]

An input-independent partition induces a Cartesian product of rooted proposer-coloured execution graphs, a direct sum of proposal-prefix antimatroids, and a blockwise product of stable matchings and their lattices.

Proof. No proposer ever reaches a receiver outside its block. Suppose some proposer in P_i made a first outside proposal; immediately before it, every receiver in R_i would have rejected it and would hold some other proposer of P_i, so |R_i| = k receivers would hold k distinct alternatives drawn from the k - 1 remaining members of P_i, which is impossible. Every update therefore changes one block coordinate, which gives the graph product and the antimatroid direct sum. Stability is block-diagonal, because a cross-block pair would create a blocking pair with an internally preferred partner. QED

Attribution. The stable-matching part of Theorem 7 is not new. For two blocks it is Definition 6.1 and Proposition 6.2 of [21], where the same separation hypothesis gives Stab(S) = Stab(S1) x Stab(S2), a product of lattices and a disjoint union of rotation posets. What is new here is the extension to partitions with any number of blocks, the uniqueness of the finest such partition (Theorem 6), and the execution-graph product together with the antimatroid direct sum, none of which concern the terminal stable set.

The three products are easiest to believe on one instance, and Fig 1 is that instance: five proposers whose only proper blocks are the first three and the last two, so that the finest partition has exactly two parts and nothing finer exists. The digraph drawn there is at the same time the Hasse diagram of the proposal-prefix antimatroid, so its being a Cartesian product is the direct-sum statement as well, and the stable set beside it is the product of the two factor stable sets. Every line of the composition table below can be read off that one picture.



(e) every composition law of Corollary 7.2, on this one instance

invariant	I_1	I_2	I	law
state count N	10	4	40	multiplicative
arcs per state E/N	1.5	1.0	2.5	additive
proposal total Q*	4	2	6	additive
complete schedules L	8	2	240	8 x 2 x C(6,4)
commutation index	2/3	1	2/3	multiplicative
stable matchings	2	2	4	multiplicative

Fig 1. One partition, three products. [PROVED in general; EXACT for the displayed instance] A five-agent instance I = I1 + I2 whose only proper blocks are {p1,p2,p3} and {p4,p5}, so the finest input-independent partition has exactly two parts. Panel (a) is the preference table, each cell carrying the pair (the proposer's rank of the receiver, the receiver's rank of the proposer); the faded cells are the cross-block pairs, which the proof of Theorem 7 shows no run ever reaches. Panels (b) and (d) are the execution digraphs of the two factors and of the whole instance, the latter drawn as ten copies of the four-state factor, one at each state of the ten-state factor: blue arcs move a proposer of the first block and orange arcs one of the second, and the picture is the Cartesian product. Because that digraph is the Hasse diagram of the proposal-prefix antimatroid, the same panel is the direct-sum

statement: the six events split 4 + 2 and a feasible prefix is a pair of feasible prefixes. Panel (c) is the stable set, whose four elements are exactly the pairs of a stable matching of each factor. Panel (e) reads every line of Corollary 7.2 off this one instance: N = 10 x 4, E/N = 1.5 + 1.0, Q* = 4 + 2, L = 8 x 2 x C(6,4), the commutation index 2/3 x 1, and the stable count 2 x 2. All six quantities are computed from the definitions by the independent implementation, not copied from the text.

Lemma 7.1 (a direct sum never splits a proposer) [PROVED; the component description EXHAUSTIVE at n = 3]

In any direct-sum factorisation of the proposal-prefix antimatroid, all events of one proposer lie in the same factor. Consequently the behavioural factorisations are exactly the partitions of the proposer set for which the reachable pointer vectors form the product of their projections.

Proof. Suppose two events of one proposer p lie in different factors. The events of p are linearly ordered by the proposal that produces them, and a proposal prefix contains an initial segment of them, so if they do not all lie in one factor then some adjacent pair is split: (p, i) in one factor and (p, i + 1) in another, with p making at least i + 2 proposals. A feasible prefix stopping p after i proposals meets the first factor in a set free of (p, i); a feasible prefix in which p makes i + 2 meets the second factor in a set containing (p, i + 1). In a direct sum the union of those two pieces is feasible, but it contains (p, i + 1) without (p, i) and is therefore not a proposal prefix. So no factorisation separates two events of one proposer, and a factorisation is determined by the induced partition of the proposers; the direct-sum condition then says exactly that the set of reachable pointer vectors is the product of its projections onto the parts.

The lemma matters twice. It makes the two decompositions of this paper commensurable: both are now partitions of agents, the input one pairing blocks of proposers with blocks of receivers and the behavioural one partitioning the proposers alone, which is why Theorem 7 holds in the direction it does and not the other. And it reduces the cost of deciding behavioural primality from the 2^{Q^*-1} bipartitions of the event set to the 2^{n-1} - 1 partitions of the proposers, which is what makes the size-four survey of a companion paper in preparation possible at all.

Corollary 7.2 (how each invariant composes, and why it matters which) [PROVED; EXHAUSTIVE at n = 3; SAMPLED at n = 4 to 6]

Over a direct sum the invariants of this paper compose in three ways and no more. Table 16 is the whole content of that monoid picture, read off one line at a time.

Table 16. Composition laws over a direct sum. [PROVED] The first five lines concern the behavioural direct sum of Lemma 7.1, a partition of the proposers; the last concerns the input partition of Theorem 7, which is the only one the stable set respects. The two-factor forms are given here; the general r-factor forms are displayed a few paragraphs below, and the column that matters is the last one. Verified with 0 failures on 260 random direct sums of sizes four to six, and for the stable set on all 888 decomposable size-three profiles.

invariant	value on I + J	kind
state count N	N(I) N(J)	multiplicative
arcs per state, E / N	E(I)/N(I) + E(J)/N(J)	additive
proposal total Q*	Q*(I) + Q*(J)	additive

invariant	value on I + J	kind
complete schedules L	the multinomial of the block totals, times L(I) L(J)	twisted
commutation index	the product of the two indices	multiplicative
stable-set size, over the input partition	the product of the two sizes	multiplicative

Proof. The Cartesian product of Theorem 7 gives the state count multiplicative and the arc count E(I)N(J) + E(J)N(I); dividing by N(I)N(J) turns that into the additivity of arcs per state, which is the form worth remembering. Every event belongs to exactly one factor, so the proposal total is additive. A complete schedule of the sum is an interleaving of one complete schedule of each factor, which gives the multinomial for L. Dividing L by the multinomial of the full terminal vector cancels the interleaving factor, so the commutation index is multiplicative; that cancellation is the reason for normalising L this way rather than any other. The last line is Theorem 7. QED

Remark (which invariants the algebra sees). The reduced types - the multisets of prime blocks - form a free commutative monoid under direct sum, which is Theorem 6 read multiplicatively, so any invariant that composes over the finest partition is fixed by its values on the primes. Table 16 sorts those invariants into the two kinds the monoid algebra distinguishes, and the sorting is what decides which of them can be enumerated. The multiplicative ones are the evaluation homomorphisms of the polynomial algebra; the additive ones are its gradings. A grading has its candidate set fixed in advance by bounds on a single prime factor, because the value on a direct sum is the sum over the parts and bounds on the parts confine the total to an interval of integers. A multiplicative invariant can be confined to a finite set the same way, so it is not that gradings alone admit a finite candidate set; what the grading gives is that the set is an interval, which is what the certifiable quotient of a companion paper in preparation enumerates. Confining a candidate set and settling every candidate in it remain different things. Of the two gradings here only the proposal total is integer-valued with both bounds proved, so it is the one that defines a certifiable quotient and arcs per state is not. That is the reason that quotient is the one it is.

The operational description given in Methods follows. The finest behavioural partition is the partition of the proposers into connected components of the graph joining p and p' when they propose to a common receiver during the run. One direction is immediate: if no two proposers on opposite sides of a partition ever address a common receiver, the two groups touch disjoint receiver sets, run independently, and their pointer vectors form a product. The converse we verified rather than proved: it holds for every one of the 46,656 size-three profiles, and for 40,000 random size-four and 8,000 random size-five profiles, with no exception. The fast test was also checked against the event-set search on all 46,656 size-three profiles, where the two agree everywhere. Call an instance doubly prime when it is prime both in the input sense and in the behavioural sense. Behavioural primeness implies input primeness, by the contrapositive of Theorem 7, so the doubly prime instances are exactly the behaviourally prime ones; there are 20,736 of them at n = 3.

With N_i reachable states, E_i edges, Q_i terminal proposals and L_i complete schedules in factor i:

$$N = \prod_i N_i, \qquad E = \sum_i E_i \prod_{j \neq i} N_j, \qquad Q^* = \sum_i Q_i^*, \qquad L = \binom{Q^*}{Q_1^*, \ldots, Q_r^*} \prod_i L_i.$$

The commutation index [PROVED, with EXHAUSTIVE calibration]

Define

$$\kappa(I) = L(I) \Big/ \binom{Q^*}{q_1^*, \ldots, q_n^*}.$$

The denominator is the number of interleavings of the per-proposer chains, that is, the number of complete executions an instance would admit if all events of different proposers were independent everywhere. So kappa measures how much the state-dependence of independence costs, and kappa = 1 exactly when the system behaves like a free shuffle of its chains. By the product formulas kappa is multiplicative over blocks, and the Boolean cube has kappa = 1. Its value is a finer invariant than the pair (N, E).

5. The converse fails, and how badly

Theorem 8 (the converse fails) [PROVED]

Behavioural decomposability does not imply input decomposability.

Proof. Take any instance whose proposers have pairwise distinct first choices. By Theorem 5 the execution graph is the Boolean cube, whose event system is the direct sum of n one-event systems, so the instance is behaviourally decomposable. The receiver orders are unconstrained, and choosing them so that no mutual top-k coalition is complemented destroys every input-independent partition. QED

EXHAUSTIVE at n = 3: 25,920 profiles are behaviourally decomposable while only 888 are input-decomposable, so 25,032 behaviourally decomposable profiles are input-prime. Conversely every behavioural block lies inside an input block, with no exceptions among the 46,656 profiles. Proposition 12.3 turns this into a named family valid at every size: the cyclic anti-phase profile C_n is input-prime while its execution is the Boolean cube, whose event system is the direct sum of n one-event systems.

The behavioural side has a uniqueness theorem of its own, and comparing the two is what shows how far apart they are.

Theorem 6.1 (the behavioural factorisation is unique too) [PROVED; EXHAUSTIVE]

Let F be any family of subsets of a finite set E. The partitions of E over which F factorises as a direct sum are closed under common refinement. Consequently F has a unique finest direct-sum factorisation, namely the meet of all of them.

Proof. F is always contained in the product of its projections, so F factorises over a partition exactly when the size of F equals the product of the projection sizes; in particular the one-block partition is always valid. Let A and B be valid partitions, and for every C in A and D in B choose an element x(C,D) of the projection of F onto C intersect D. Fix C. For each D pick a witness X(C,D) in F whose restriction to C intersect D is x(C,D). Because B is valid, assembling the D-parts of these witnesses gives an element U of F with U restricted to D equal to X(C,D) restricted to D for every D; hence U restricted to C is the disjoint union over D of the x(C,D), and that union lies in the projection of F onto C. Because A is valid, assembling these C-parts gives the disjoint union over all C and D of the x(C,D) as an element of F. So F factorises over the common refinement of A and B. The meet of the finitely many valid partitions is therefore itself valid, and it refines every valid partition. QED

Corollary 6.2 (unique prime decomposition). Both decompositions used in this paper are unique. On the input side the finest partition is unique by Theorem 6 and its blocks are input-prime; on the behavioural side the finest direct-sum factorisation of the proposal-prefix family is unique by Theorem 6.1 and its blocks are behaviourally prime. In either case, if an instance is written as a composition of factors that are all prime, that decomposition is the finest one, so the factors are determined by the instance - not merely up to isomorphism, but as a partition of the labelled agent set. No ordering convention is needed.

EXHAUSTIVE support. Over all 255 non-empty set families on three events and all 65,535 on four events, no pair of valid partitions has an invalid meet and every family has a valid finest factorisation. Over all 46,656 size-three instances the same holds for the proposal-prefix family, and the finest

factorisation never splits one proposer's events between blocks. Its block sizes are 1+1+1 for the 10,368 Boolean cubes, 1+3 for 15,552 instances, and a single block of 5, 6 or 7 events for 14,256, 5,832 and 648 instances respectively - which is the event-level form of the behavioural counts.

Remark 6.3 (commutativity is not the mechanism). It is tempting to argue that, because composition of independent blocks is commutative and associative, uniqueness follows from the definition of a prime instance alone. It does not. Commutativity and associativity together with irreducibility are not sufficient for unique factorisation: in the numerical semigroup generated by 2 and 3 under addition, both generators are irreducible, yet $6 = 2 + 2 + 2 = 3 + 3$ has two factorisations into irreducibles, of different lengths. What is needed is the stronger property that the valid decompositions are closed under common refinement - Theorem 6 on the input side, Theorem 6.1 on the behavioural side. The blocks here are prime in that strong sense, not merely irreducible, and it is the strong sense that the proofs use.

Remark 6.4 (an imposed order is a normal form; a derived order can be more). Uniqueness also cannot be obtained by imposing an order on the factors of a commutative decomposition. Such an ordering replaces a multiset of factors by a sequence, which removes the ambiguity of which factor is written first but cannot identify two genuinely different factorisations: sorting the two factorisations of 6 above still leaves (2,2,2) and (3,3). Where factors are sorted in this paper - by block size and then by the canonical form of the sub-instance - that is only to obtain a canonical sequence for hashing and for the certificates in the supplementary material. An order derived from the structure rather than imposed on it is a different matter, and the execution space carries exactly such an order.

Remark 6.5 (where a non-commutative composition would enter). A non-commutative composition exists nearby and is not covered by these theorems. Require only the proposer half of the block condition: every proposer of P_1 ranks all of R_1 above everything outside R_1, with no condition on the receivers of R_1. This relation is not symmetric in the two blocks, and it admits no product theorem. Witness: the instance in which all six agents rank the opposite side in the identical order. Proposer p_1 has top choice r_0, so the pair ({p_1},{r_0}) satisfies the proposer half; but r_0 ranks p_0 first, the instance is behaviourally prime, and its execution has 15 states and 6 terminal proposals, so nothing factors. The one-sided condition is also nearly vacuous as a decomposition criterion: every instance satisfies it for some singleton pair outright, since any proposer paired with its own first choice will do, and 46,272 of the 46,656 size-three instances satisfy it for some singleton pair that is not itself a block, including all 20,736 behaviourally prime ones; the remaining 384 are those in which every proposer is mutually first-matched. Both directions of dominance are therefore needed, and it is the symmetric condition that makes the composition commutative in the first place.

Proposition 9.10 (the dynamic decomposition structure is flat, and its counting degenerates). [PROVED; EXHAUSTIVE at n = 3; EXACT at n = 4 to 6] (a) The behavioural factorisations of an instance are exactly the coarsenings of the unique finest one. So if the finest has r parts the instance has exactly B(r) of them, B the Bell number, and that count depends on nothing else about the instance.

(b) By Lemma 7.1 the finest factorisation is a partition of the proposers, so the dynamic decomposition structure is a partition of n and not a forest: there is no nesting to record.

(c) Every partition of n is the shape of the finest factorisation of some input-prime instance at n = 3, 4, 5 and 6, so at those sizes the dynamic decomposition-level classification is complete, with exactly p(n) classes: 3, 5, 7 and 11.

Proof. (a) Theorem 6.1 makes the factorisations closed under common refinement, so a finest one exists and is unique. If s is any factorisation, the meet of s with the finest is a factorisation at or below the finest, hence equal to it, so the finest refines s. Conversely merging factors of a direct sum leaves a direct sum, so every coarsening of the finest is a factorisation. The factorisations are therefore in bijection with the set partitions of the r parts, and there are B(r) of those. (b) is Lemma 7.1. (c) by exhibiting an instance for each partition. QED

Verification. Part (a) was checked by brute force, enumerating every partition of the proposers and testing the product condition directly, over all 46,656 size-three profiles and 4,800 random size-four and size-five ones. The count was B(r) every time, the pairs of the number of prime factors and the number of factorisations taking only the values (1, 1), (2, 2), (3, 5), (4, 15) and (5, 52).

The two families below are the sharpest separation this paper has, and the symmetry needed to describe them is short enough to give here.

An instance automorphism is a pair g = (g_P, g_R) of permutations of the two sides that carries the ranked bipartite graph to itself. Aut(I) is defined as a subgroup of S_n x S_n and deliberately excludes the exchange of the two sides; side exchange is treated separately below.

Lemma 10 (rigidity) - recalled from [21, Lemma 4.1]

If g in Aut(I) fixes one agent then g is the identity.

Proof. Suppose g_P(p_0) = p_0. Preservation of p_0's strict complete order over all receivers forces g_R to be the identity. With g_R trivial, preservation of every receiver's strict complete order over all proposers forces g_P to be the identity. The same argument runs with the sides interchanged. Completeness of the lists is used in both steps. QED

Proposition 11 (free action and divisibility) - recalled from [21, Proposition 4.2], semiregularity from [39]

Aut(I) acts freely on P and on R. Consequently |Aut(I)| divides n, every non-identity automorphism is fixed-point-free on both sides, and every orbit has size |Aut(I)|.

Lemma 10 and Proposition 11 are stated for self-containedness and are not claimed as contributions. They are Lemma 4.1 and Proposition 4.2 of [21], which itself identifies Proposition 11 as the profile-level specialisation of a semiregularity property of preference-profile stabilisers proved in [39]. Following [21, Definition 4.4] we call a profile automorphism-maximal when its automorphism group has order n, that is when Aut(I) acts regularly on each side; [21, Theorem 4.5] puts every such profile in a regular normal form over a group of order n.

Everything this paper derives from symmetry flows from Proposition 11. For odd n a side-preserving involution is impossible. For prime n = p, Aut(I) is trivial or cyclic of order p acting regularly, so non-trivial and automorphism-maximal coincide at prime size. The coincidence does not survive to n = 4, where a group of order two is non-trivial without being maximal. Non-abelian automorphism groups first become possible at n = 6, since a group acting freely on at most five points has order 1, 2, 3, 4 or 5 and is abelian. What is new below is none of this, but the passage from these constraints to the execution digraph, and the exact counting of symmetric instances.

Theorem 12 (regular symmetry forces the Boolean cube) [PROVED]

If |Aut(I)| = n, so that Aut(I) acts regularly on each side, then G(I) is the n-dimensional Boolean cube.

Proof. Write f(p) for the first choice of p. Equivariance gives f(g_P(p)) = g_R(f(p)). Since the action on P is simply transitive, f is determined by its value at one proposer, and since the action on R is simply transitive, f is a bijection. The first choices are therefore pairwise distinct, and Theorem 5 applies. QED

The converse is false: Boolean-cube execution does not require any input symmetry at all, as the next subsection quantifies.

Corollary 12.1 (the regular anti-phase templates run as a Boolean cube) [PROVED; EXHAUSTIVE for orders at most 5]

Every automorphism-maximal profile has Boolean-cube execution. In particular so does every regular anti-phase template P(G, A) of [21], for every finite group G and every ordering A.

Proof. In the regular normal form P(G, A, B) of [21, Theorem 4.5] the first choice of the proposer indexed by g is the receiver indexed by g a_0, and g maps to g a_0 is a bijection of G. First choices are therefore pairwise distinct and Theorem 5 applies. QED

So on any such instance deferred acceptance makes exactly n proposals, none of which is rejected, and stops at the matching g maps to g a_0. The execution invariants are forced: $N = 2^n$, $E = n\,2^{n-1}$, Q* = n, L = n! and kappa = 1.

Table 4a. Regular anti-phase templates, execution against stable set. Computed directly; the stable-set column agrees with [21, Observation 7.2].

G	normalised orderings	N	E	Q*	L	kappa	size of Stab
Z2	1	4	4	2	2	1	2
Z3	2	8	12	3	6	1	3
Z4	6	16	32	4	24	1	4, 6 or 8
V4	6	16	32	4	24	1	10
Z5	24	32	80	5	120	1	5

Proposition 12.2 (execution structure and the stable set are independent invariants) [PROVED, with certificates; EXHAUSTIVE at n = 3]

The proposer-coloured execution digraph does not determine the number of stable matchings, and the number of stable matchings does not determine the execution digraph.

Certificates. All six normalised orderings of the cyclic group of order four give templates with the identical execution digraph of Table 4a, while their stable sets have 4, 6 and 8 elements [21, Observation 7.2]; this proves the first half. The Klein template has the same execution digraph again and ten stable matchings, the classical maximum f(4) at size four [21, Observation 5.13], the maximum itself being known to grow simply exponentially [13]; so the size-four instance with the largest stable set has the smallest execution graph a size-four instance can have. For the second half, an exhaustive pass over all 46,656 size-three profiles shows that every value of the stable-matching count occurs in several distinct execution classes.

Proposition 12.3 (two Latin families that the execution cannot tell apart) [PROVED; EXHAUSTIVE for n at most 6]

The sharpest form of the separation uses two families that exist at every size. Both are rank-Latin and automorphism-maximal, so by Corollary 12.1 both run as the Boolean cube; they sit at opposite extremes of everything the execution digraph cannot see. A strict-Latin instance has no proper block at all, at any size, and for a short reason: a proposer's first choice lies in its own block, and the Latin condition makes that receiver rank the proposer n-th, so a block containing the pair must have size at least n. It is prime in the third sense too, and for the reason that makes the Latin instances interesting in the first place: for each t the matching pairing every proposer with its t-th choice is stable, because a blocking pair would need ranks summing to less than n + 1. That is the stability criterion of Benjamin, Converse and Krieger discussed below, and the author restated it as [37, Theorem 2]. The n rank-matchings therefore exhaust K_(n,n) between them, the union of the stable matchings is complete, and the finest partition in the third sense is the trivial one. C_n is therefore prime in the input sense and in the third sense at once, and at the same time it runs as the Boolean cube, which is the most decomposable behaviour there is: its n proposers make one proposal each, to n distinct receivers, so the event system splits into n single-event factors. That is the separation of Theorem 8 in its sharpest form. The statement is about C_n and the strict-Latin condition; A_n is rank-Latin but not strict-Latin, and sits at the opposite end of the input axis. The four-agent Latin instance of [38, Example 8] is decomposed there in the blocking-diagram sense of that paper, which is a condition on neither the input nor the stable set and is not compared with either here. Latin instances in the strict sense, where the rank of a receiver in a proposer's list and of that proposer in the receiver's list sum to n + 1, are the latin marriages of Benjamin, Converse and Krieger [14] and go back to Knuth [19]. Their stability criterion forces at least n stable matchings on any such instance, and their family S_n attains that minimum; C_n is S_n up to relabelling, which is a second derivation of the count in the table. A_n has a single stable matching at every size and so cannot be Latin in that sense at all; it is rank-Latin only in the weaker sense used here, where each rank level is a perfect matching on each side. [EXHAUSTIVE for n = 2 to 6]

The aligned cyclic Latin profile A_n gives the receiver indexed by j rank 1 + ((j - i) mod n) in the list of the proposer indexed by i, and symmetrically on the other side, so every pair (p_i, r_i) is mutual first. It is the profile introduced in the companion submission on Latin rank balance. The cyclic anti-phase profile C_n has the proposer indexed by i rank the receivers from i upwards and the receiver indexed by j rank the proposers from j+1 upwards; it is the profile of [21, Section 3], and it has no mutual-first pair at all.

For every n at least 2, A_n and C_n have the same reachable execution digraph, namely the Boolean cube with $N = 2^n$, $E = n\ 2^{n-1}$, Q = n, L = n! and kappa = 1, and both have automorphism group of order n. Their finest input partitions are the all-singletons partition and the one-block partition respectively, so A_n is maximally input-decomposable and C_n is input-prime; and their stable sets have 1 and n elements respectively.*

Table 4d. The two cyclic Latin families. [EXHAUSTIVE for n = 2 to 6; the execution row follows from Corollary 12.1 for every n]

	A_n, aligned	C_n, anti-phase
rank-Latin	yes	yes
order of Aut	n	n

	A_n, aligned	C_n, anti-phase
execution digraph	Boolean cube B_n	Boolean cube B_n
commutation index kappa	1	1
mutual-first pairs	n, a perfect matching	0
finest input partition	1+1+…+1	one block, input-prime
size of the stable set	1	n

Proof. Both profiles are invariant under simultaneous rotation of the two sides, so each has automorphism group of order at least n, and Proposition 11 forces equality; Corollary 12.1 then gives the Boolean cube and its invariants. In A_n the pair (p_i, r_i) ranks each other first for every i, so the all-singletons partition is input-independent and, being the finest possible, is the finest; the stable set is a single matching because a perfect matching of mutual-first pairs is forced. In C_n the top choice of the proposer indexed by i is the receiver indexed by i, whose own top choice is the proposer indexed by i+1, so no block of size one exists; for $1 < k < n$ the top k receivers of the proposer indexed by i are the interval from i to i+k-1, which moves with i, so no block of size k exists either and C_n is input-prime. Its stable set has n elements [21, Proposition 3.3]. QED

This one family settles three things at once and at every size, where the earlier certificates settled them only at $n = 4$. The execution digraph does not determine the stable-matching count, since A_n and C_n share it and differ by a factor of n. It does not determine input decomposability either, which sharpens Theorem 8 from an existence statement to a named family: C_n is input-prime with fully parallel behaviour. And rank-Latinness, a strong input regularity, constrains none of it - the point the companion submission makes for the stable set, holding here for the execution as well.

6. The exact census

Theorem 9 (exact census of input-prime instances) [PROVED; EXACT]

Let T(n) = (n!)^2n be the number of balanced strict complete instances of size n, let Pi(n) be the number of input-prime ones, and for a partition lambda = (k_1, ..., k_s) of n with multiplicities m_j write N_lambda(n) for the number of paired block structures of that shape. Then Pi(1) = 1 and

$$T(n) = \sum_{\lambda=(k_1,\dots,k_s)\vdash n} N_\lambda(n) \prod_{i=1}^{s} \Pi(k_i)\big((n-k_i)!\big)^{2k_i}, \qquad N_\lambda(n) = \frac{1}{\prod_{j\geq 1} m_j(\lambda)!} \left(\frac{n!}{\prod_i k_i!}\right)^2.$$

The lambda-term counts exactly the instances whose finest partition has shape lambda, and the s = 1 term is Pi(n), so the identity determines Pi(n) recursively.

Proof. By Theorem 6 the finest partition is unique, so the instances are partitioned by its shape. Fix a block structure with blocks (P_i, R_i) of sizes k_i. A proposer in P_i has a list consisting of an arbitrary order of R_i followed by an arbitrary order of its complement: k_i! times (n - k_i)! choices, of which the first factor is exactly the datum of the restricted sub-instance and the second is free. The same holds on the receiver side. The structure is the finest partition precisely when each restricted sub-instance is input-prime, since the finest partition of the whole instance is the union of the finest partitions of its blocks. Collecting the free tails gives ((n - k_i)!)^2k_i per block and the restricted data gives Pi(k_i). Counting the block structures of shape lambda gives N_lambda(n). QED

Table 3. Exact census of input-prime instances.

n	instances T(n)	input-decomposable	input-prime Pi(n)	decomposable fraction
1	1	0	1	0
2	16	2	14	1.250e-1
3	46,656	888	45,768	1.903e-2
4	110,075,314,176	88,478,208	109,986,835,968	8.038e-4
5	6.1917e20	11,557,648,579,461,120	6.1916e20	1.867e-5
6	1.9408e34	7.8739e27	1.9408e34	4.057e-7
7	6.8238e51	5.7129e43	6.8238e51	8.372e-9
8	4.8790e73	7.5836e63	4.8790e73	1.554e-10

The n = 3 row reproduces the exhaustive enumeration, and the n = 4 row was confirmed independently by the side-splitting enumeration described in Methods. At n = 4 the finest partition has shape 3+1 for 26,362,368 profiles, 2+2 for 903,168, 2+1+1 for 20,901,888 and 1+1+1+1 for 40,310,784. The size-five row is exact too: T(5) = 619,173,642,240,000,000,000 and Pi(5) = 619,162,084,591,420,538,880, abbreviated in the table only to keep the column narrow.

Remark 9.3 (blocks of size one are mutual-first pairs). An input-independent block of size one is a pair that ranks each other first. Such pairs are the mutual-first pairs of the companion submission on Latin rank balance, where they are shown to be pairwise disjoint and to belong to every stable matching, so that a perfect matching of them certifies a unique stable matching. In the language used here that is the extreme case of Theorem 7: when the finest partition is all singletons, every factor

carries a single proposal event, the execution graph is the Boolean cube, and the stable-matching lattice is a product of one-element lattices. Theorem 9 counts those instances exactly. The all-singletons term is n! times $((n-1)!)^{2n}$ - that is 2 at n = 2, 384 at n = 3 and 40,310,784 at n = 4 - so the forced-pair criterion applies to a fraction $n!/n^{2n}$ of all instances, and by Corollary 9.1 that single term dominates input decomposability as n grows.

What the present framework adds is the case k at least 2. A mutual-first pair is a mutual top-one coalition; an input block of size k is a mutual top-k coalition. Theorem 6 shows that these are laminar for all k simultaneously, which is why a finest partition exists and is unique, and that in turn is what makes the product formulas of Theorem 7 and the census of Theorem 9 possible. None of it follows from the size-one case, and conversely nothing here supplies the certificate theory for the stable set that the companion submission develops.

Corollary 9.1 (asymptotics, with the leading correction). [PROVED; EXACT at n = 2 to 12] Write W(lambda) for the lambda-term of the census identity. Then $W(1^n)$ is exactly the fraction $n!/n^{2n}$ of T(n); the shape $(2, 1^{n-2})$ contributes exactly $7n/(2(n-1)^3)$ of $W(1^n)$, and the shape $(2, 2, 1^{n-4})$ exactly $49n(n-2)(n-3)/(8(n-1)^7)$; and all remaining shapes with at least two parts together contribute $O(n^{-6})$ of it. The one-part shape lambda = (n) is the prime term Pi(n) itself and is not part of D(n), so the quantifier here and the bound below range over the shapes with every part at most n - 1. Hence

$D(n)/T(n) = (n!/n^{2n})\ (1 + 7n/(2(n-1)^3) + O(n^{-4})) = (n!/n^{2n})(1 + \mathrm{Theta}(n^{-2}))$.

Proof. For lambda $= 1^n$ the multiplicity divisor is n!, so $N = (n!)^2/n! = n!$, and each block contributes $\mathrm{Pi}(1)((n-1)!)^2$; hence $W(1^n) = n!((n-1)!)^{2n}$, which against $T(n) = (n!)^{2n}$ is $n!/n^{2n}$. For lambda $= (2, 1^{n-2})$ the divisor is (n-2)!, giving $N = (n!)^2/(4(n-2)!)$, while the blocks contribute $\mathrm{Pi}(2)((n-2)!)^4 = 14((n-2)!)^4$ and $((n-1)!)^{2(n-2)}$; dividing by $W(1^n)$ and using $(n-1)! = (n-1)(n-2)!$ and $n! = n(n-1)(n-2)!$ leaves $7n/(2(n-1)^3)$, the 7/2 being Pi(2)/4. The same computation with two blocks of size two, whose divisor carries an extra 2!, gives $49n(n-2)(n-3)/(8(n-1)^7)$.

For the remaining shapes bound Pi(k) by $T(k) = (k!)^{2k}$ and put $g(k) = k!(n-k)!/(n-1)!$, which is k/C(n-1,k-1), so that g(1) = 1 and g(k) <= 1 throughout 1 <= k <= n-1. Replacing the falling factorial n!/(n-K)! by n^K, where K is the number of agents lying in blocks of size at least two, gives

$W(\mathrm{lambda})/W(1^n) <= (1/\mathrm{prod_}(j>=2)\ m_j!)\ \mathrm{prod_}(k_i >= 2)\ \mathrm{phi}(k_i)$, $\mathrm{phi}(k) = n^k\ g(k)^{2k}/(k!)^2$,

and summing over every shape is then at most exp(Phi) - 1 with Phi = sum_(k >= 2) phi(k), because the multiplicity divisors are exactly the ones that exponential supplies. Here $\mathrm{phi}(2) = 4n^2/(n-1)^4 = O(n^{-2})$. For k >= 3 the estimate $C(n-1,k-1) >= (n-k+1)^{k-1}/(k-1)!$ gives $g(k) <= k!/(n-k+1)^{k-1}$ and hence $\mathrm{phi}(k) <= n^{k(3-2k)}(k!)^{2k-2}\ 4^{k(k-1)}$ while k <= n/2, which is $O(n^{-9})$ at k = 3 and smaller beyond; for k > n/2 the bound g(k) <= 1 against $(k!)^2$ is already superexponentially small. So Phi $= \mathrm{phi}(2) + O(n^{-9})$. The shapes other than 1^n and $(2, 1^{n-2})$ either carry a block of size at least three, contributing at most exp(Phi) sum_(k >= 3) phi(k) = $O(n^{-9})$, or carry at least two blocks of size two, contributing at most the tail of the exponential from m = 2 onwards, which is $O(\mathrm{Phi}^2) = O(n^{-4})$. QED

Verification. The recursion was run in exact integer arithmetic to n = 12 by asymp.py. At every size the census identity of Theorem 9 closes; the all-singletons share equals $n!/n^{2n}$ exactly; and the two closed forms above hold exactly, the first giving 20,901,888/40,310,784 at n = 4, which is the ratio of the 2+1+1 count to the 1+1+1+1 count reported above. After the three leading shapes are removed, n^6 times the remaining share falls to 8.3 at n = 12 and is still decreasing, against the 7.15

that the third term of the series predicts.

The ratio of the true fraction to $n!/n^{2n}$ is 2.195 at n = 4, 1.519 at n = 5, 1.227 at n = 6, 1.127 at n = 7, 1.063 at n = 9 and 1.032 at n = 12. Input decomposability is therefore vanishingly rare, and randomly sampled profiles are a poor way to study it: at n = 4 one expects 8.04 decomposable profiles per 10,000 draws.

Corollary 9.2 (which multisets of factors occur). Let p(m) be the number of input-prime isomorphism classes of size m. The number b(n) of multisets of prime classes realisable as the factorisation of a size-n instance is the Euler transform of p:

$$\sum_{n\geq 0} b(n)x^n = \prod_{m\geq 1} (1-x^m)^{-p(m)}.$$

Since p(1) = 1, every partition of n contributes, and the multiset of prime factors together with their classes is a complete invariant of the finest decomposition. [PROVED, given Theorem 6]

Corollary 13 (census for prime size) [PROVED; EXACT]

For prime p the number of instances with a non-trivial automorphism is

$$A(p) = \frac{\big((p-1)!\big)^2\,(p!)^2}{p-1}.$$

The count of pairs of p-cycles is $((p\text{-}1)!)^2$, each such pair fixes $(p!)^2$ instances, and each symmetric instance is fixed by p - 1 non-identity automorphisms. This gives 72 at p = 3 and 2,073,600 at p = 5.

The recursion behind the census

The census of Theorem 9 is one instance of a general reduction, and the reduction is what makes the same question answerable at the next size. Hall's marriage condition has a shape worth imitating here.

Hall's marriage condition has a shape worth imitating: it is quantified over all subsets of one side, so a single sentence covers every size at once, size-specific quantities can then be chased by recurrence, and induction on n is the natural proof method. This section asks what that shape buys here. It buys a sharper statement of the block criterion, an exact reduction of every extremal problem to input-prime instances, and a replacement for the paper's warning about the two senses of the word prime. It does not buy the values themselves, and one pattern that the framework invites turns out to be false.

Hall's condition is vacuous here; its tight sets are not [PROVED; EXHAUSTIVE at n = 3]

With strict complete lists the underlying graph is the complete bipartite graph, so the Hall inequality holds for every subset and a perfect matching exists for free. Hall's theorem therefore contributes nothing directly. What does carry content is the tight-set half of Hall theory, applied not to the complete graph but to the levels M_k of the mutual-rank filtration. Call a set A of proposers tight

at level k when |A| = k, its M_k-neighbourhood B satisfies |B| = k, and the M_k-neighbourhood of B is A again.

Proposition 21 (a block is tight and regular, and tightness alone is not enough) [PROVED; EXHAUSTIVE at n = 3]

Every input-independent block is tight at its level. The converse is false: tightness does not force the k-regularity that Theorem 9.4 requires. Among the 46,656 size-three profiles, 16,896 carry a set that is tight at some level without being a block, as do 11,089 of 20,000 random size-four profiles.

Certificate. Proposers p1: r1 r2 r3, p2: r1 r2 r3, p3: r1 r3 r2; receivers r1: p1 p2 p3, r2: p1 p3 p2, r3: p1 p2 p3. At level 2 the set {p1, p2} has M_2-neighbourhood {r1, r2}, so the Hall inequality is tight with |N(A)| = |A| = 2. But p2 has M_2-degree 1, because r2 ranks p3 above p2, so the level-2 subgraph is not 2-regular and {p1, p2} is not a block. The blocks of this instance are the mutual-first pair ({p1}; {r1}) and the whole instance.

So the Hall-shaped form of Theorem 9.4 reads: (A, B) is a block exactly when A is tight at level k = |A| and the induced level-k subgraph is k-regular. Regularity is not decoration, and the comparison with Hall theory has to be made carefully. Tight sets in the classical sense, those with |N(A)| = |A| and no condition on the size, are closed under union and intersection when Hall's condition holds, which is why they are usually described as a lattice; but M_k has maximum degree k, so for k < n it need not satisfy Hall's condition and the closure can fail. In the profile where all six agents rank the opposite side in the same order, {p1, p2} and {p1, p3} are both tight in M_2 while their union has only two neighbours. The sets tight at a fixed level are not closed under either operation for a second and simpler reason: both change the size, and the definition above fixes it at k. What survives of the comparison is the part that matters: tight sets, at a level or not, can cross, while blocks are laminar (Proposition 9.5), which is strictly stronger. The gap is visible at once: among the size-three profiles, 768 carry a crossing pair of tight sets, 2,304 such pairs in total, while no instance at either size carries a crossing pair of blocks. A witness is p1: r1 r2 r3, p2: r1 r3 r2, p3: r2 r3 r1 with r1: p1 p3 p2, r2: p2 p3 p1, r3: p1 p2 p3, where {p1, p2} and {p1, p3} are both tight at level 2 and cross. Regularity is exactly what upgrades the family of tight sets to a forest.

Theorem 21.1 (the partition recursion) [PROVED]

Let phi be an instance invariant that composes over the finest input partition under an associative commutative operation that is monotone in each argument: Q* composes by addition, and N, the number of reachable partial matchings, |Stab| and the commutation index compose by multiplication, and all four are non-negative, which is what makes the composition monotone. Monotonicity is a hypothesis and not a consequence of the others: for a multiplicative invariant allowed to take negative values the maximum over a product is not the product of the maxima. Write phi-circle(k) for the maximum of phi over input-prime instances of size k. Then for every n

$$\max_{|I|=n} \phi(I) \;=\; \max_{\lambda \vdash n} \bigcirc_i \, \phi^{\circ}(\lambda_i),$$

the maximum running over partitions of n and the composition over the parts. The same reduction applied to counting instead of maximising is Theorem 9.

Proof. Theorem 7 gives the composition rule over any input-independent partition, and Theorem 6 makes the finest one unique, with prime blocks. Since the composition is monotone in each argument, replacing each factor by the maximum over prime instances of its size can only increase the value, which yields the inequality in one direction. For the other, given prime instances of sizes k_1 to k_s, place them in disjoint blocks and let every agent rank its own block above everything outside, using the block's own order inside and any order outside; the blocks are then input-independent, and since each is prime the finest partition is exactly that one. So every partition is realised and the maximum is attained.

Proposition 21.4 (how large the state count of an indecomposable instance must be) [PROVED; EXHAUSTIVE at n = 2 and 3]

Let I be doubly prime of size n, in the sense defined in Section 4, with terminal pointer vector q* and Q* proposals in total. The hypothesis cannot be weakened to input primeness: the two-agent instance of Section 2 is input-prime and has Q* = 2. Then

$$Q^* \geq 2n-1, \qquad 2^n + Q^* - n \leq N \leq \prod_p (q_p^* + 1) \leq \left(n + \tfrac{1}{n}\right)^n.$$

In particular N is at least 2^n + n - 1, so the state count of an indecomposable instance is exponential in the size and no polynomial expression in n exists.

Proof. Every proposer ends matched, so all n receivers are addressed; write d(r) for the number of proposals receiver r receives, so that Q* is the sum of the d(r). By Lemma 7.1 the graph joining two proposers that address a common receiver is connected, and its edges are covered by one clique per receiver, the clique at r having d(r) vertices. A clique on d(r) vertices supplies at most d(r) - 1 edges of a spanning tree, so the sum of the d(r) - 1 is at least n - 1, and Q* is that sum plus the number of addressed receivers, which is n. For the lower bound on N, every 0/1 pointer vector is feasible: let each proposer in a chosen subset propose exactly once. That is 2^n states. The all-ones vector is feasible and lies below q*, and the run continues from it to q* in Q* - n further proposals, each giving a distinct feasible vector with some coordinate at least 2 and hence outside the cube. The upper bound is the grid the feasible vectors lie in, then the arithmetic-geometric mean inequality, then the classical bound Q* at most n^2 - n + 1.

The bound on Q* is attained at every size checked: the smallest totals are 3, 5, 7, 9 and 11 at n = 2 to 6, which is 2n - 1 exactly. The bound on N is attained only at n = 2. The smallest state counts are 5 and 11 at n = 2 and 3 exhaustively, and constrained search finds 23, 47 and 95 at n = 4, 5 and 6, which are upper bounds on those minima; all five equal 3 x 2^{n-1} - 1, which is open problem A of Section 7. At the other end the largest state counts found on indecomposable instances are 5, 15, 52 and 232 at n = 2 to 5. The first three are the unconstrained maxima at those sizes; at n = 5 the unconstrained maximum is known only to be at least 232, which is open problem B, so there the statement is that an indecomposable instance attains the largest value known. Indecomposable instances therefore attain the extreme in both directions wherever the extreme is known.

7. Open problems

Five, chosen so that each can be attacked at the next size by a reader with a computer, and so that each has data behind it rather than only a hunch.

A. Is the least state count of a doubly prime instance exactly $3 \times 2^{n-1} - 1$? Proposition 21.4 proves only $2^n + n - 1$. The measured minima are 5, 11, 23, 47 and 95 at $n = 2$ to 6, exhaustively at $n = 2$ and 3 and by constrained search above, and all five equal $3 \times 2^{n-1} - 1$. Every minimiser found also attains the proposal floor $Q^* = 2n - 1$, so a second question is whether the two minima are always achieved together.

B. What is the exact maximum number of reachable states? It is 2, 5, 15 and 52 at $n = 1$ to 4 and at least 232 at $n = 5$. The first four values are the Bell numbers $B(2)$ to $B(5)$, which is what makes the problem interesting: the pattern fails at $n = 5$, where $B(6) = 203$ is already exceeded. A closed form, or a proof that none of the obvious ones holds, would settle the extremal side of the theory.

C. Is the proposal-total spectrum always a full interval? For a doubly prime instance the proposal total lies in $[2n - 1, n^2 - n + 1]$, the lower bound by Proposition 21.4 and the upper by the classical bound. Every value in that interval is attained at $n = 3$, 4, 5 and 6, by census at $n = 3$ and by exhibited certificates above. Is that true at every size?

D. Is every block-forest shape realised? Up to relabelling a block forest is a rooted tree whose root has size n and whose children have sizes below their parent summing to at most it. There are $f(n)$ such shapes, with f from $n = 1$ to 7 equal to 1, 3, 10, 40, 170, 785 and 3,770, and exactly $2\,f(n - 1)$ of them are open to a prime instance. Every shape is realised at $n = 2$ to 6. The construction that realises them fails only through accidental blocks, so the question is whether some shape forces one at a larger size.

E. Does double primality cost anything except at size four? The largest stable set over doubly prime instances is 3 at $n = 3$, which is the unrestricted maximum; 6 at $n = 4$ against an unrestricted 10; 16 at $n = 5$, which is again the unrestricted maximum; and 26 at $n = 6$, which is also the best the unrestricted search reached. Size four is so far the only size where the constraint is known to cost anything at all. Is it the only one?

Data and code availability

The verification code accompanies this paper as ancillary files, in the anc/ directory of the arXiv submission, and the full repository is at https://github.com/i4dagh/smprime, archived at doi:10.5281/zenodo.22984730. It contains two implementations of the reachable-state construction written independently of each other from the definitions, the scripts behind every count reported here as EXHAUSTIVE or EXACT, the certificates for the statements tagged TARGETED, and the script that draws Fig 1 from the definitions rather than from the text. Every exhaustive count in this paper was produced twice, once by each implementation, and the two agreed. The archived version is the one to cite.

Generative AI disclosure

Large language models (OpenAI ChatGPT, Google Gemini and Anthropic Claude) were used while this manuscript was prepared: to organise claims, to edit language, and to propose and run verification checks. Every mathematical statement, proof, computation and citation was reviewed by the author, who takes responsibility for the content.

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